Uber Revenue Per Kilometer Optimization

Through Identifying low performing trip segments

Introduction:

Uber Technologies operates a high-volume ride-hailing network across a major metropolitan city, processing thousands of trips daily across diverse geographic, temporal, and operational conditions. Despite robust trip volume, certain segments generate materially lower Revenue per Kilometer (RPK), eroding overall network profitability. The primary objective is to quantify RPK performance drivers and deliver targeted, data-backed recommendations for pricing, surge policy, and operational optimization.

Objective:

The objective of this analysis is to evaluate Uber's trip-level performance to identify segments that contribute to low Revenue per Kilometer (RPK) and recommend data-driven strategies for improvement. By examining how factors such as trip distance, time of day, weather, traffic conditions, and vehicle quality influence fare efficiency, the study aims to uncover operational and pricing inefficiencies within the city's ride network. The ultimate goal is to enhance revenue productivity, optimize supply allocation, and improve overall profitability through actionable insights.

Business Impact :

Uber's trip performance using Revenue per Kilometer (RPK) to identify revenue inefficiencies across ride segments. Despite strong overall trip volumes, short-distance trips, off-peak hours, weekends, and adverse weather conditions consistently deliver lower RPK, indicating that a large share of demand is concentrated in low-yield segments.

The impact on the business is driven by suboptimal unit economics. Fixed and semi-variable costs—including driver incentives, fuel-related expenses, and platform operations—remain largely constant across trips, causing margins to compress when RPK declines. As a result, operational resources and driver supply are disproportionately allocated to trips that generate lower revenue efficiency.

To improve performance, the analysis recommends strategic pricing and supply optimization, including minimum fare adjustments for short trips, refined incentive structures during low-RPK periods, and supply prioritization during peak and high-distance time windows. Weather-based pricing adjustments can further protect margins without materially affecting demand.

Even modest improvements in RPK across high-volume underperforming segments can drive significant revenue uplift at scale. These insights enable leadership to balance growth with profitability, improve driver earnings sustainability, and strengthen long-term marketplace efficiency.

Data Set Overview:

Data set Name: Uber Analytical dataset

Number of Rows: 482675

Number of coloumns:30

Description: trip-level Uber ride data capturing distance, fare, time-based attributes, and contextual factors (weekend, peak hours, weather) to analyze Revenue per Kilometer (RPK) and identify revenue efficiency across ride segments.

Data Columns Information:

user_id — Unique identifier assigned to each rider.

user_name — Name of the rider who booked the trip.

driver_name – Name of the driver who completed the trip.

car_condition – Condition rating of the vehicle used for the ride.

weather – Weather situation during the trip (sunny, rainy, etc.

traffic_condition – Traffic level observed during the trip.

key – Unique trip ID for identifying each ride.

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fare_amount — Total fare charged to the rider for the trip.

pickup_datetime — Exact timestamp when the trip started.

pickup_longitude – Longitude of the pickup location.

pickup_latitude — Latitude of the pickup location.

dropoff_longitude — Longitude of the dropoff location.

dropoff_latitude — Latitude of the dropoff location.

passenger_count — Number of passengers on the trip.

hour — Hour of day extracted from pickup time.

day — Day of the month extracted from pickup time.

month — Month extracted from pickup time.

weekday — Day of the week extracted from pickup time.

year — Year extracted from pickup time.

distance — Total distance traveled for the trip.

jfk_dist — Distance from pickup point to JFK Airport.

ewr_dist — Distance from pickup point to Newark Airport (EWR).

lga_dist — Distance from pickup point to LaGuardia Airport (LGA).

sol_dist – Distance from pickup point to a major landmark in South Manhattan.

nyc_dist – Distance from pickup point to NYC city center.

bearing – Direction of travel from pickup to dropoff

Data cleaning and preparation:

Data Validation and Integrity Checks

- Reviewed dataset structure to ensure each row represented a unique Uber trip.
- Identified and removed duplicate records to prevent inflation of trip volume and revenue metrics.

Handling Missing and Invalid Values

- Examined key numerical fields, including fare amount and trip distance, for missing or null values.
- Excluded records with zero or negative fare or distance values, as these distort revenue efficiency and RPK .

Outlier Detection and Treatment

- Conducted outlier analysis on fare amount and trip distance using the Interquartile Range (IQR) method.
- Capped or removed extreme values beyond acceptable bounds to reduce skewness while preserving overall data distribution.

Standardization of Categorical Variables

- Cleaned and standardized categorical fields such as weekday, weekend indicator, peak-hour flag, and weather condition to ensure consistent labeling and grouping.
- Validated category completeness to avoid misclassification during segmentation analysis.

Time-Based Feature Engineering

- Transformed raw time variables into meaningful analytical dimensions, including hour of day and peak versus off-peak periods.
- Ensured correct weekday sequencing to support trend and comparison analysis.

Metric Derivation

- Calculated Revenue per Kilometer (RPK) as the core performance metric by dividing fare amount by trip distance.
- Validated RPK distribution to confirm logical consistency across trip segments.

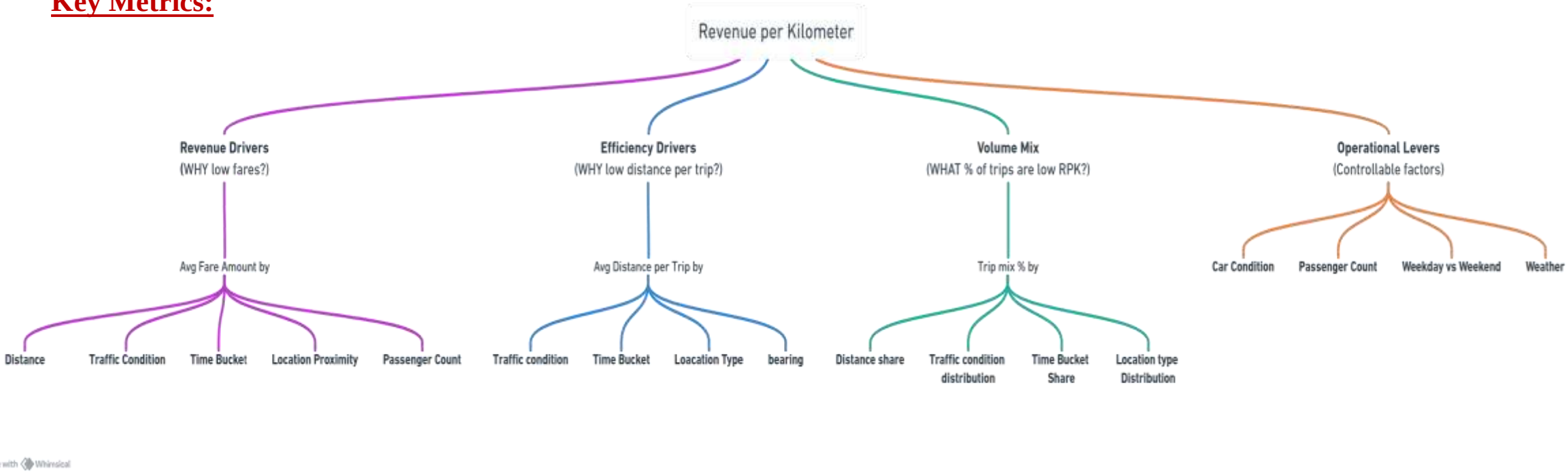
Segmentation Preparation

- Created distance buckets to enable comparative analysis across short, medium, and long trips.
- Verified segment balance to ensure interpretability of insights.

Final Dataset Readiness

- Performed final sanity checks on aggregated metrics and distributions.
- Prepared a clean, structured dataset optimized for reliable analysis, visualization, and strategic business decision-making.

Key Metrics:



Exploratory Data Analysis :

Exploratory Data Analysis (EDA) is a critical step in understanding the structure, patterns, and underlying drivers within the dataset before conducting deeper analysis or modeling. EDA is used to examine how Revenue per Kilometer (RPK) varies across different trip characteristics, including distance, time of day, day of week, and external conditions such as weather. Detailed analysis of the dataset is given below:

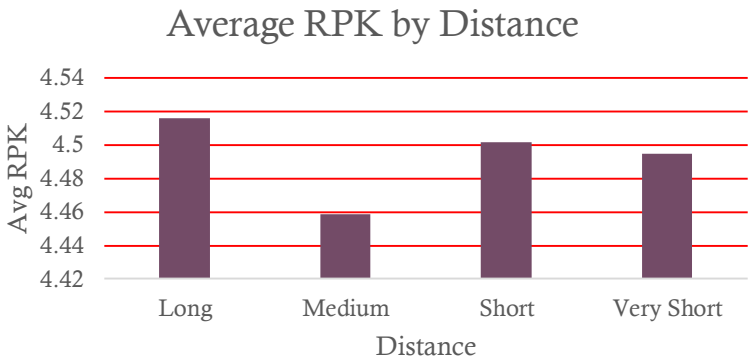
Hypothesis 1: Impact of Trip Distance on Revenue per Kilometer

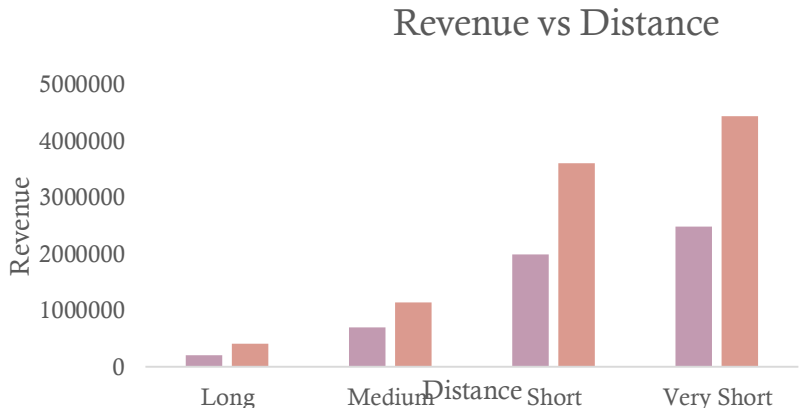
Key Findings:

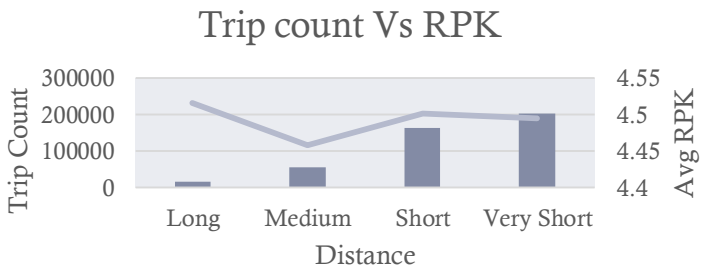
Hypothesis - Imapct of Trip Distance on Revenue per kilometer.

Analysis- RPK varies slightly across distance segments, with long trips generating the highest average RPK and medium trips generating the lowest,short and very short trips maintain RPK levels close to the overall average but dominate total revenue due to very high trip volumes.Medium distance trips emerge as the weakest segment because they combine lower RPK with moderate volume,while long trips despite strong unit economics, contribute minimally to total revenue due to low freuency .

Conclusion -The hypothesis is partially accepted, as trip distance does influence RPK but not in a strictly linear or dominant manner. While Long trips are the most efficient on a per-kilometer basis, Short and Very Short trips remain the primary revenue drivers, and medium trips represent the key area of revenue leakage. This implies that improving pricing and operational efficiency in the Medium-distance segment will deliver the greatest impact on overall revenue performance, while maintaining volume stability in Short trips and selectively scaling Long trips for margin improvement.







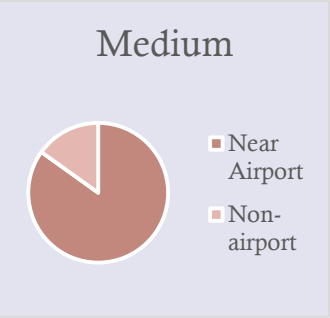
Hypothesis 2: Trips near Airports generate higher Revenue per kilometer

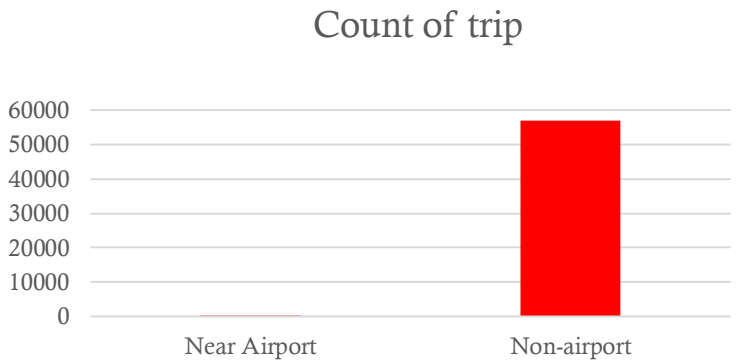
Key Findings

Hypothesis: Trips near Airports generate higher Revenue per kilometer.

Analysis: There is a sharp contrast between airport and non-airport trips. Although airport-related trips account for a very small share of total volume compared to non-airport trips (over 446,000 trips), their average RPK is significantly higher across all distance buckets. Near-airport trips show exceptionally high RPK values for long, medium, short, and very short distances, while non-airport trips remain consistently low regardless of distance. This indicates that airport rides benefit from premium pricing factors such as surge demand, fixed airport fees, longer minimum fares, and customers' lower price sensitivity. In contrast, non-airport trips dominate volume but suffer from compressed pricing, competitive alternatives, and shorter urban distances, leading to low revenue efficiency per kilometer.

Conclusion : Airport trips, despite their low frequency, are disproportionately high-value and contribute far more revenue efficiency per kilometer than non-airport trips. First, protect and optimize the airport segment through better driver availability, reduced cancellations, and targeted incentives; second, improve non-airport RPK by introducing pricing optimization for short urban trips, bundling fees, or demand-based micro-surge models.





Hypothesis 3 : City proximity and trip distance jointly influence RPK

City/distance	Long	Medium	Short	Very Short
Core	5.74	5.84	6.07	5.98
Mid	4.46	4.42	4.46	4.43
Outer	4.48	4.41	4.45	4.46

Key Findings

Hypothesis : City proximity and trip distance jointly influence RPK.

Analysis: Core city trips consistently generate the highest RPK across all distance buckets, ranging from 5.74 for long trips to a peak of 6.07 for short trips. This indicates strong pricing efficiency in dense urban areas where demand is high and idle time is low. In contrast, Mid and Outer city zones show significantly lower and relatively flat RPK values across all distance categories. Notably, reducing trip distance does not meaningfully improve RPK in Mid and Outer zones, suggesting that distance-based pricing advantages are offset by lower demand density, longer pickup times, or reduced fare premiums outside the city core.

Conclusion: city proximity and trip distance together influence RPK, but city proximity is the dominant driver. Core city areas outperform Mid and Outer zones regardless of trip length, while shorter trips only meaningfully improve RPK within the Core. Uber should prioritize driver supply and short-trip incentives in Core city zones to maximize RPK, while re-evaluating pricing, minimum fares, or pickup compensation in Mid and Outer zones to prevent persistently low revenue efficiency.

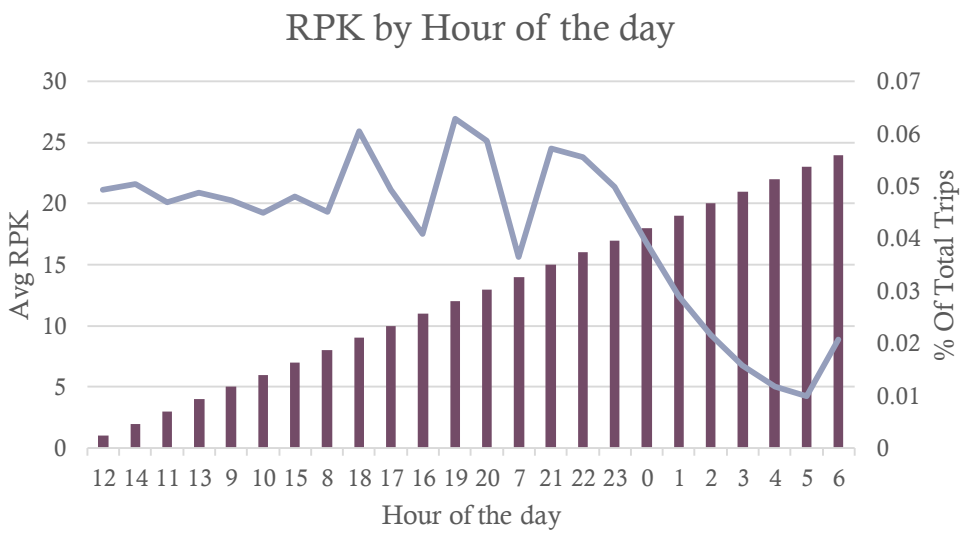
Hypothesis 4: RPK varies significantly by hour of the day ,with late night and erly morning trips generating lower RPK compared to peak daytime and evening hours

Key Findings

Hypothesis: RPK varies significantly by hour of the day ,with late night and early morning trips generating lower RPK compared to peak daytime and evening hours .

Analysis: Revenue per Kilometer remains relatively strong during daytime and evening hours, driven by higher demand density, shorter urban trips, and more effective dynamic pricing during commute and business travel periods. In contrast, RPK drops significantly during late-night and early-morning hours, even when trips still occur. This decline is likely caused by longer average trip distances, limited surge pricing, and demand-stimulation discounts that reduce fare realization per kilometer. Additionally, lower trip concentration during these hours reduces pricing power, making off-peak periods structurally less profitable on an RPK basis despite operational availability.

Conclusion : hour-of-day is a critical driver of RPK performance. Late-night and early-morning hours are structurally low-RPK segments, negatively impacting overall profitability despite operational continuity. Uber should consider time-based pricing optimization, targeted surcharges for low-supply hours, or incentives to encourage shorter, higher-margin trips during off-peak periods.



Hypothesis 5: RPK varies across weekdays and time buckets with early morning and night consistently generating low RPK Compared to afternoon, evening and morning period.

Average RPK by Weekday and time bucket

Weekday/ time bucket	Night	Morning	Evening	Early Morning	Afternoon
0	3.86	4.62	4.35	3.85	4.78
1	4.06	4.68	4.55	3.80	4.99
2	4.04	4.79	4.60	3.79	5.19
3	4.10	4.73	4.64	3.87	5.08
4	4.28	4.65	4.63	3.99	4.97
5	4.38	4.08	4.58	4.01	4.55
6	3.96	4.08	4.26	4.04	4.40

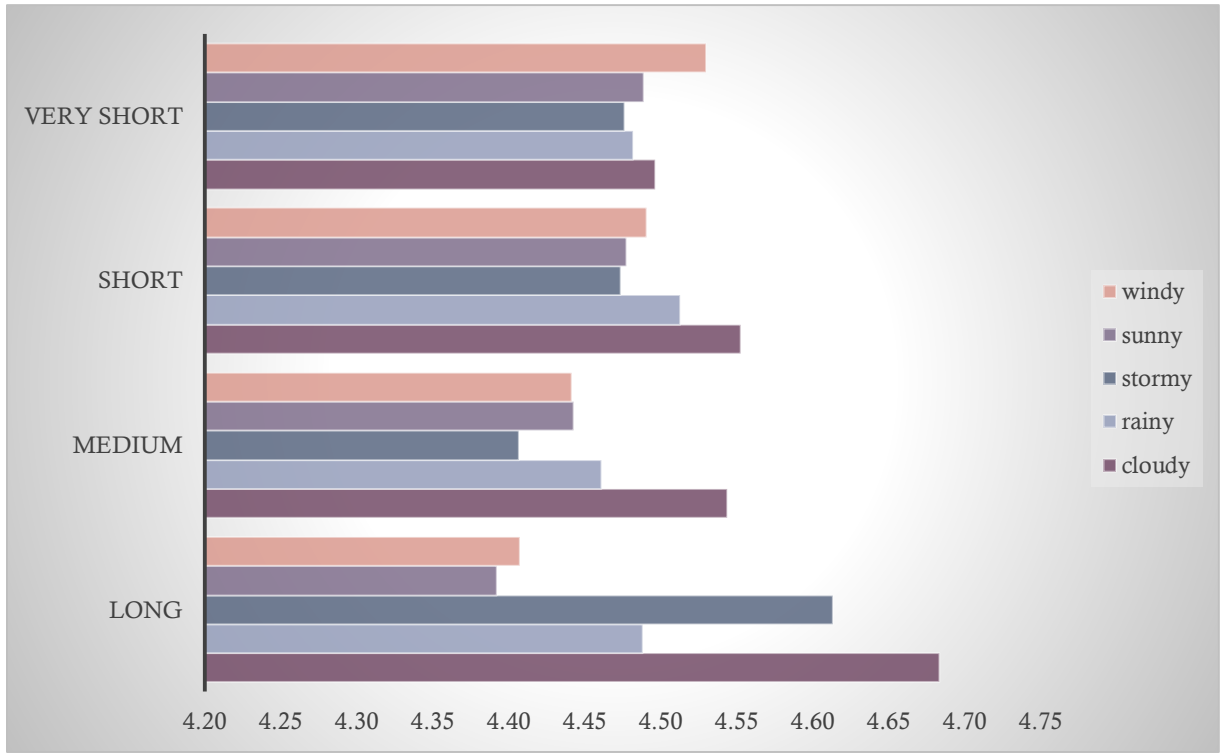
Key Findings

Hypothesis: RPK varies across weekdays and time buckets with early morning and night consistently generating lower RPK compared to afternoon ,evening and morning period.

Analysis : Afternoon and Morning time buckets consistently deliver the highest RPK across most weekdays, reflecting strong demand density, shorter urban trips, and effective fare realization. Evening RPK remains stable but slightly lower, suggesting moderate congestion or competitive pricing effects. In contrast, Early Morning and Night time buckets show visibly lower RPK across all weekdays, with Early Morning being the weakest . This pattern implies longer trip distances, limited surge pricing, and discount-driven demand during off-peak hours. The consistency of low RPK across weekdays confirms that time-of-day, rather than day-of-week, is the dominant driver of RPK performance.

Conclusion : RPK is significantly influenced by time buckets, with Early Morning and Night periods structurally underperforming regardless of weekday. This insights suggest opportunities for time-based pricing adjustments, minimum fare enforcement, or targeted incentives focused on improving RPK during off-peak hours, while continuing to maximize supply and surge efficiency during high-performing afternoon and morning periods to protect overall revenue efficiency.

Hypothesis 6: weather conditions impact RPK differently across trip distance categories



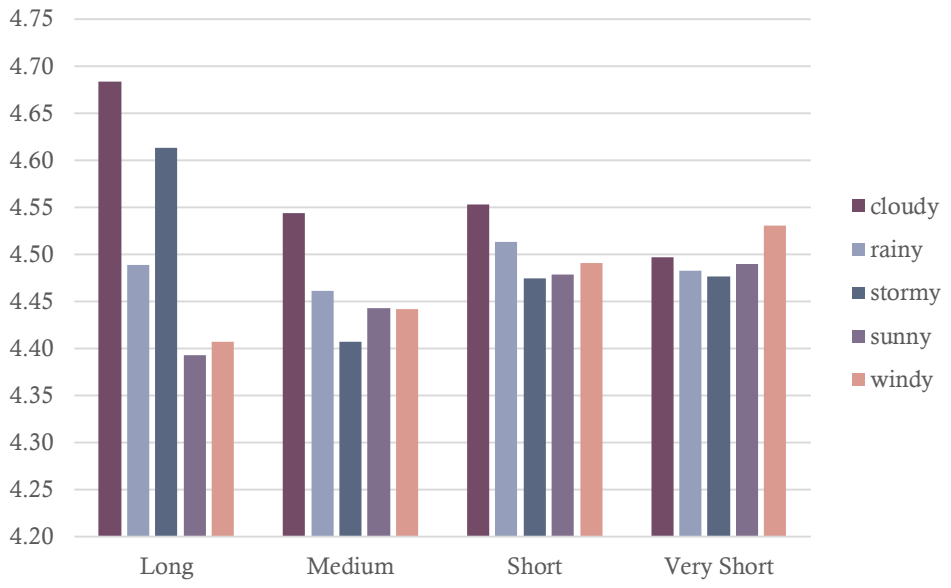
Key Findings

Hypothesis: weather conditions impact RPK differently across trip distance categories

Analysis -It compares average RPK across four trip distance buckets (Very Short, Short, Medium, Long) under different weather conditions (cloudy, rainy, stormy, sunny, windy). Overall, RPK remains relatively stable across weather types, ranging narrowly , indicating limited aggregate weather sensitivity. However, when segmented by distance, clear patterns emerge. Long-distance trips show the highest RPK during cloudy and stormy conditions, suggesting higher fares or surge pricing during adverse weather for longer rides. In contrast, Very Short trips show slightly better RPK in windy conditions, while rainy weather tends to suppress RPK marginally across most distance categories. Medium and Short trips exhibit minimal variation across weather types, indicating pricing consistency or capped fare adjustments for these segments.

Conclusion- While weather does not significantly affect overall RPK at an aggregate level, its impact varies meaningfully by trip distance. Adverse weather conditions positively influence RPK primarily for long-distance trips, likely due to increased demand, reduced supply, or surge pricing effects. For shorter trips, weather has minimal influence on revenue efficiency. Uber can improve profitability by enabling more dynamic, distance-sensitive pricing strategies during adverse weather, particularly for long-distance rides, while maintaining stable pricing for short and medium trips to preserve rider trust and volume .

Hypothesis 7: Trip distance has a stronger impact on RPK than weather condition



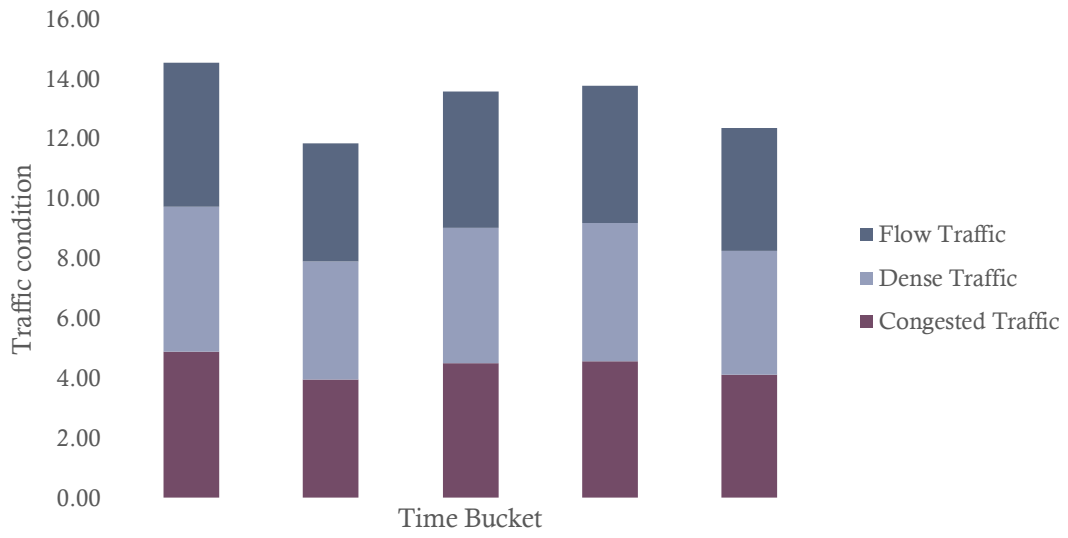
Key Findings

Hypothesis: Trip distance has a stronger impact on RPK than weather condition.

Analysis: The data compares average RPK across four distance categories (Very Short, Short, Medium, Long) under five weather conditions (cloudy, rainy, stormy, sunny, windy). Across all weather types, long-distance trips consistently generate higher RPK values, particularly under cloudy (4.68) and stormy (4.61) conditions. In contrast, medium and short trips show relatively compressed RPK values with minimal fluctuation across weather types, indicating pricing stability. Very short trips exhibit slightly higher RPK during windy conditions (4.53), but the variation remains marginal. Overall, the spread in RPK across distance categories is more pronounced than the spread across weather conditions, suggesting that distance is the primary driver of revenue efficiency rather than weather alone.

Conclusion: While weather conditions cause minor fluctuations in RPK, trip distance plays a more significant and consistent role in determining revenue efficiency. Long-distance trips benefit the most from adverse or cloudy weather, likely due to surge pricing and higher fare accumulation per kilometer. Shorter trips remain relatively insulated from weather-driven pricing changes. Uber should prioritize distance-based pricing optimization and selectively apply weather-based surge multipliers to long-distance trips to maximize RPK without impacting rider trust on shorter journeys.

Hypothesis 8: Traffic conditions and time of day jointly influence RPK



Key Findings

Hypothesis : Traffic conditions and time of the day jointly influence RPK.

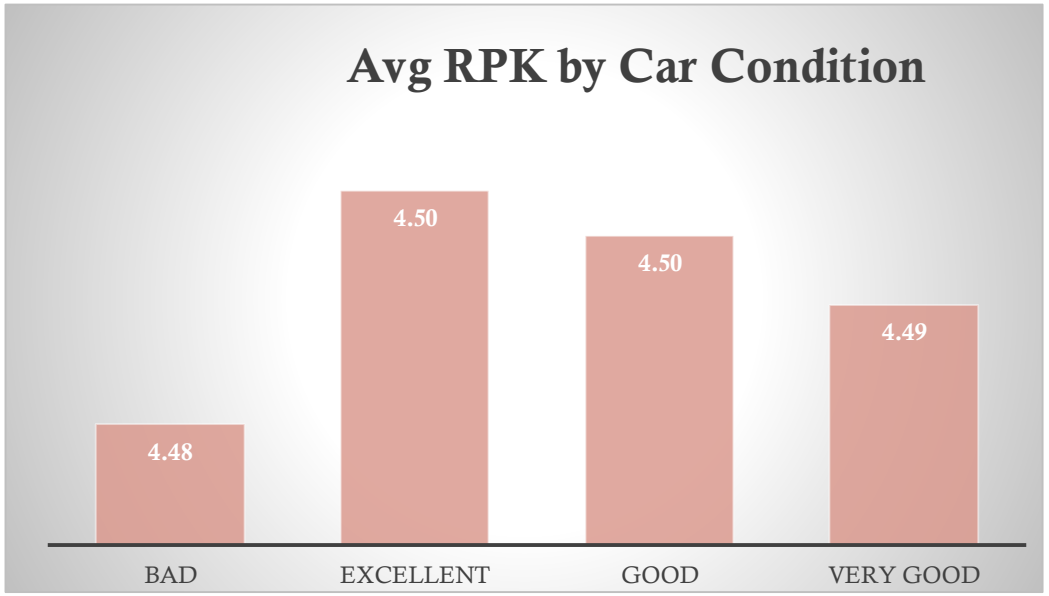
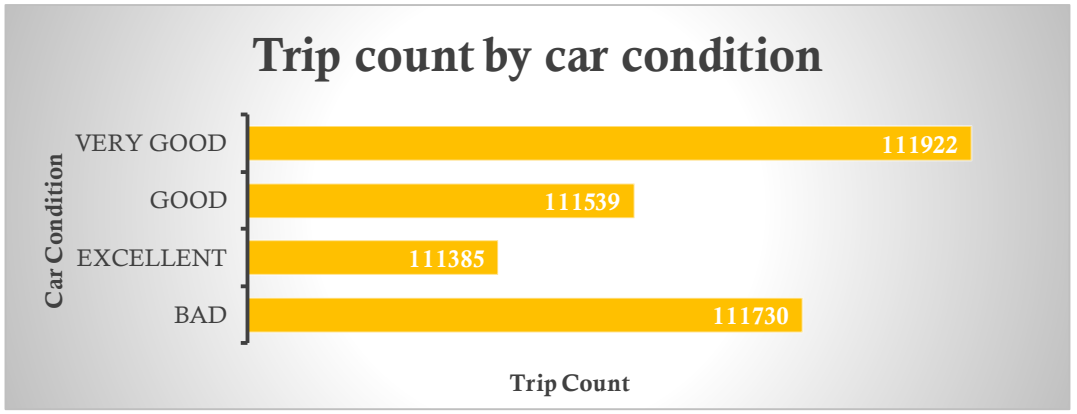
Analysis: The data indicates that both time of day and traffic conditions collectively impact RPK, though the variation across traffic types is relatively moderate. Afternoon trips generate the highest RPK across all traffic conditions, peaking under congested traffic , followed closely by dense and flow traffic. Morning and evening periods also show strong RPK values, suggesting consistent demand and fare efficiency during core commuting hours. In contrast, early morning trips record the lowest RPK across all traffic types , likely due to lower fares, promotional pricing, or reduced demand. Interestingly, congested traffic marginally outperforms flow traffic in most time slots, indicating that higher fares during congestion may compensate for longer travel times.

Conclusion: As RPK is clearly influenced by the interaction between traffic conditions and time of day. Peak daytime and commuting hours deliver stronger revenue efficiency, particularly during congested conditions where fare multipliers offset delays. Early morning and night trips consistently underperform, signaling opportunities for targeted pricing strategies, demand stimulation, or cost optimization. Uber can improve overall RPK by prioritizing driver availability and surge optimization during afternoon, morning, and evening peak periods while reassessing incentives and pricing structures for low-yield early morning segments.

Hypothesis 9: Car Condition Influences RPK and Trip Volume

Key Findings

Hypothesis : Car Condition Influences RPK and trip volume

Analysis: The a average RPK remains largely consistent across vehicles rated as Excellent (4.50), Good (4.50), and Very Good (4.49), while cars in Bad condition record a slightly lower RPK of 4.48. This indicates that vehicle condition has only a marginal impact on revenue efficiency per kilometer. However, trip volume differs more noticeably across conditions. Vehicles in Very Good condition generate the highest number of trips (111,922), followed by Bad (111,730), Good (111,539), and Excellent (111,385). The relatively high trip count for better-rated vehicles suggests higher rider preference or availability, even though fare efficiency per kilometer remains stable. Overall, customer demand appears more sensitive to car condition than pricing efficiency.

Conclusion: While car condition does not significantly influence RPK, it does affect trip volume. Maintaining vehicles in Good or Very Good condition helps sustain higher utilization and rider preference, even if per-kilometer revenue remains unchanged. Uber should continue enforcing vehicle quality standards to drive higher trip frequency rather than expecting improvements in RPK. Investments in driver vehicle quality programs will therefore contribute more to demand growth and customer satisfaction than direct revenue efficiency gains.